

1. EC2 Instance

Definition

EC2 (Elastic Compute Cloud) is a web service provided by **Amazon Web Services (AWS)** that allows users to create and run virtual servers (called **instances**) in the cloud. It provides scalable computing capacity, so you can quickly launch, manage, and terminate virtual machines without buying physical hardware.

Uses of EC2

- Host websites and web applications.
- Run databases.
- Software development and testing.
- Big data processing.
- Machine Learning workloads.
- Gaming servers.

Advantages

- Pay only for the resources you use.
 - Easily increase or decrease capacity.
 - High availability.
 - Secure environment.
 - Easy integration with other AWS services.
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2. Features of EC2

1. Scalability

- Easily increase or decrease the number of instances based on demand.

2. Elasticity

- Resources automatically adjust according to workload.

3. Pay-as-you-Go

- Pay only for the compute time you use.

4. Multiple Operating Systems

- Supports Linux, Windows, Ubuntu, Red Hat, Amazon Linux, etc.

5. High Availability

- Deploy instances across multiple Availability Zones.

6. Security

- Uses Security Groups, IAM roles, and Key Pairs for secure access.

7. Storage Integration

- Can be attached to Amazon EBS volumes for persistent storage.

8. Monitoring

- Integrated with Amazon CloudWatch for monitoring CPU, memory, and network usage.
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3. EC2 Instance Types

AWS provides different instance types based on workload requirements.

A. General Purpose

- Balanced CPU, memory, and networking.
- Suitable for web servers, small databases, and business applications.

Examples: T3, T4g, M5, M6i

B. Compute Optimized

- High CPU performance.
- Suitable for scientific computing, gaming, and batch processing.

Examples: C5, C6i

C. Memory Optimized

- High RAM capacity.
- Suitable for in-memory databases, real-time analytics, and caching.

Examples: R5, R6i, X2

D. Storage Optimized

- High-speed local storage.
- Suitable for big data, NoSQL databases, and data warehousing.

Examples: I3, D2

E. Accelerated Computing

- Includes GPU or FPGA.
- Suitable for Machine Learning, AI, graphics rendering, and video processing.

Examples: P4, G5, F1

4. Key Pair

Definition

A **Key Pair** is a security mechanism used to securely connect to an EC2 instance. It consists of:

- **Public Key** – Stored by AWS on the EC2 instance.
- **Private Key** – Downloaded and stored securely by the user.

When connecting:

- AWS checks the public key.
 - You use the private key to authenticate.
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Types of Key Pairs

1. RSA Key Pair

- Most commonly used.
- Supports SSH access.
- Compatible with Linux and Windows.

2. ED25519 Key Pair

- Modern cryptographic algorithm.
 - Faster and more secure than RSA.
 - Used mainly for Linux instances.
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Benefits of Key Pairs

- Secure authentication.
 - Password-less login.
 - Strong encryption.
 - Prevents unauthorized access.
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5. Security Group

Definition

A **Security Group** is a **virtual firewall** that controls **inbound (incoming)** and **outbound (outgoing)** traffic for an EC2 instance.

It allows only the traffic you specify.

Features of Security Groups

1. Instance-level firewall.
 2. Stateful (return traffic is automatically allowed).
 3. Controls inbound and outbound traffic.
 4. Allows only permitted traffic.
 5. Multiple security groups can be attached to one instance.
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Security Group Rules

Inbound Rules

Control traffic entering the EC2 instance.

Example:

Protocol Port Purpose

SSH	22	Remote login
HTTP	80	Web traffic
HTTPS	443	Secure web traffic

Outbound Rules

Control traffic leaving the EC2 instance.

By default, all outbound traffic is allowed.

Benefits of Security Groups

- Protects EC2 instances from unauthorized access.
 - Controls network traffic.
 - Improves security.
 - Easy to modify without restarting the instance.
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Difference Between Key Pair and Security Group

Key Pair	Security Group
Used for authentication	Used for network security
Allows secure login	Controls incoming/outgoing traffic
Public and private key	Firewall rules
Required for SSH/RDP access	Required for port access

- **EC2 Instance:** An EC2 instance is a virtual server in AWS used to run applications in the cloud.
- **Features of EC2:** Scalability, elasticity, pay-as-you-go pricing, multiple operating systems, high availability, security, storage integration, and monitoring.
- **Instance Types:** General Purpose, Compute Optimized, Memory Optimized, Storage Optimized, and Accelerated Computing.
- **Key Pair:** A pair of public and private keys used for secure authentication to an EC2 instance.
- **Security Group:** A virtual firewall that controls inbound and outbound network traffic for an EC2 instance

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